TSM 099

Homework 6

Linear and quadratic equations

Topics 24-28 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact answers. For equations with fractions, combine and simplify each side first. Then move each remaining denominator to multiply the entire opposite side. Check solutions in the original equation.

Linear equations and fractions

1. Solve and check.

$$-\frac{3}{5}x = 9$$

2. Solve and check.

$$4(2x - 3) - 3(x + 1) = 2x + 6$$

3. Solve each equation; state the full solution set.

$$2(3x - 1) + 5 = 6x + 3$$

$$5(x + 1) = 5x + 2$$

4. Combine and simplify each side first, then solve and check.

$$\frac{x+1}{3} + \frac{x-2}{6} = \frac{5}{2}$$

5. Combine and simplify each side first, then solve and check.

$$\frac{2x-1}{4} = \frac{x+2}{3}$$

Three quadratic methods

6. Solve by factoring.

$$x^2 + 3x = 18$$

7. Solve by factoring.

$$2x^2 - 7x + 3 = 0$$

8. Solve by taking square roots.

$$2(x + 1)^2 = 18$$

9. Solve by taking square roots. Give exact real solutions.

$$(x - 3)^2 = 7$$

10. Solve with the quadratic formula.

$$2x^2 + x - 4 = 0$$

11. Solve with the quadratic formula.

$$3x^2 - 2x = 2$$

12. Solve over the real numbers. Then state the complex solutions if complex numbers are allowed.

$$x^2 + 4 = 0$$

Choose and explain

13. Choose an efficient method and solve. Explain the method choice.

$$5(x - 2)^2 = 20$$

14. A student solves x squared - 5x = 0 by dividing by x. Explain the lost solution and solve correctly.